

Finger Print Recognition for Home Security Using IOT

Thank you for the opportunity to explain my project.

Project Overview

The project I'd like to present is a Biometric Home Security System using Fingerprint Recognition. The primary objective is to design and implement a secure, convenient, and cost-effective door lock system that utilizes fingerprint verification for authentication.

Problem Statement

Traditional door locks can be compromised using duplicate keys or guessed passwords. This project aims to overcome these limitations by leveraging unique biometric characteristics, specifically fingerprints, to ensure enhanced security.

System Components

The system consists of:

1. Hardware:
 - RR307 Fingerprint sensor
 - NODE MCU
 - Relay Module
 - Solenoid Lock
 - 12V DC adapter
 - Breadboard
2. Software:
 - Arduino IDE
 - Adafruit Fingerprint library

Procedure:

Hardware Setup:

1. Connect the RR307 Fingerprint sensor to the NODE MCU.
2. Connect the Relay Module to the NODE MCU and the Solenoid Lock.
3. Connect the 12V DC adapter to the system.

Here's an explanation of why each connection is made in the hardware setup:

1. Connecting RR307 Fingerprint sensor to NODE MCU:

- The RR307 Fingerprint sensor is connected to the NODE MCU to enable fingerprint recognition and verification.
- The NODE MCU reads fingerprint data from the sensor and processes it for authentication.
- This connection allows the system to identify and verify individuals based on their fingerprints.

2. Connecting Relay Module to NODE MCU and Solenoid Lock:

- The Relay Module acts as a switch to control the Solenoid Lock.

- The NODE MCU sends signals to the Relay Module to activate or deactivate the Solenoid Lock.
- When a fingerprint is verified, the NODE MCU triggers the Relay Module to activate the Solenoid Lock, unlocking it.

3. Connecting 12V DC adapter to the system:

- The 12V DC adapter provides power to the entire system.
- The NODE MCU, Relay Module, Solenoid Lock, and Fingerprint sensor all require power to function.
- The adapter ensures a stable and reliable power supply.

In summary, this setup creates a fingerprint-based security system:

1. Fingerprint recognition and verification
2. Lock control through the Relay Module
3. Power supply for all components

This system can be used for various applications, such as secure door locks, cabinets, or access control systems.

Software Procedure:

Fingerprint Registration Code:

```
C
#include <SoftwareSerial.h>
#include <Adafruit_Fingerprint.h>

// Define pins for serial communication
#define RX_PIN 4
#define TX_PIN 5

// Create a SoftwareSerial object
SoftwareSerial mySerial(RX_PIN, TX_PIN);

// Create an Adafruit_Fingerprint object
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

void setup() {
  Serial.begin(9600);
  // Initialize fingerprint sensor
  finger.begin(57600);
  // Check if fingerprint sensor is found
  if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
  } else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) {
      delay(1);
    }
  }
}

void loop() {
```

```

// Enroll a new fingerprint
Serial.println("Ready to enroll a fingerprint!");
// Read ID number from user
uint8_t id = readnumber();
// Enroll fingerprint
getFingerprintEnroll(id);
}

uint8_t getFingerprintEnroll(uint8_t id) {
// Capture fingerprint image
int p = -1;
Serial.print("Waiting for valid finger to enroll as #");
Serial.println(id);
while (p != FINGERPRINT_OK) {
p = finger.getImage();
switch (p) {
case FINGERPRINT_OK:
Serial.println("Image taken");
break;
// Handle errors
}
}
// Convert image to template
p = finger.image2Tz(1);
// Store template in database
finger.storeModel(id);
}

```

Fingerprint Recognition Code:

```

C
#include <Adafruit_Fingerprint.h>

// Define pins for serial communication
#if (defined(__AVR__) || defined(ESP8266)) && !defined(__AVR_ATmega2560__)
SoftwareSerial mySerial(4, 5);
#define RELAY_PIN D5
#else
#define mySerial Serial1
#endif

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

void setup() {
Serial.begin(9600);
// Initialize fingerprint sensor
finger.begin(57600);
// Check if fingerprint sensor is found
if (finger.verifyPassword()) {
Serial.println("Found fingerprint sensor!");
} else {
Serial.println("Did not find fingerprint sensor :(");
while (1) {
delay(1);
}
}
// Initialize relay pin
pinMode(RELAY_PIN, OUTPUT);
digitalWrite(RELAY_PIN, HIGH);
}

```

```

void loop() {
  // Get fingerprint ID
  getFingerprintID();
  delay(50);
}

uint8_t getFingerprintID() {
  // Capture fingerprint image
  uint8_t p = finger.getImage();
  // Convert image to template
  p = finger.image2Tz();
  // Search for matching template
  p = finger.fingerSearch();
  if (p == FINGERPRINT_OK) {
    // Unlock door
    digitalWrite(RELAY_PIN, LOW);
    delay(ACCESS_DELAY);
    digitalWrite(RELAY_PIN, HIGH);
  }
}

```

Explanation of Key Functions:

1. `getFingerprintEnroll()`: Captures fingerprint image, converts it to a template, and stores it in the database.
2. `getFingerprintID()`: Captures fingerprint image, converts it to a template, and searches for a matching template in the database.
3. `finger.getImage()`: Captures fingerprint image from sensor.
4. `finger.image2Tz()`: Converts fingerprint image to template.
5. `finger.storeModel()`: Stores template in database.
6. `finger.fingerSearch()`: Searches for matching template in database.

Libraries Used:

1. `Adafruit_Fingerprint`: Provides functions for interacting with fingerprint sensor.
2. `SoftwareSerial`: Enables serial communication on non-standard pins.
3. The `getFingerprintEnroll()` function is used to enroll new fingerprints, while the `getFingerprintID()` function is used to verify existing fingerprints.
4. The `finger.begin()` function initializes the fingerprint sensor, and `finger.verifyPassword()` checks if the sensor is properly connected.
5. The `Serial.begin()` function initializes serial communication for debugging purposes.

Flowcharts:

Fingerprint Enrollment:

1. Start
2. Initialize fingerprint sensor
3. Check if sensor is found
4. Read ID number from user
5. Capture fingerprint image

6. Convert image to template
7. Store template in database
8. End

Fingerprint Verification:

1. Start
2. Initialize fingerprint sensor
3. Check if sensor is found
4. Capture fingerprint image
5. Convert image to template
6. Search for matching template
7. Unlock door if match found
8. End

Troubleshooting:

1. Fingerprint sensor not detected: Check connections, ensure proper wiring.
2. Enrollment fails: Ensure proper finger placement, check sensor calibration.
3. Verification fails: Check template matching algorithm, ensure sufficient database storage.

Secure Login and Threat Prevention in Bank Management System:

Project title- "Secure Login & Threat Prevention In BankManagement System"

This project aims to enhance the security of a bank management website by implementing strong password policies, encrypted passwords, and additional authentication measures.

These measures help mitigate the risks of brute force attacks, SQL injection, cross-site scripting (XSS), and unauthorized access.

The project focuses on improving the security and integrity of sensitive financial data, ensuring a safer online banking experience for users.

Experimental setup

- Database management system (DBMS) software -WAMPSEVER (phpmyadmin)
- Programming language - PHP,JAVASCRIPT,HTML,CSS.
- Database Management – MySQL
- Integrated Development Environment (IDE) for coding and testing - Visual Studio Code

Introduction

The security of online banking systems is crucial to protect users' financial information from malicious attacks. This project addresses this need by implementing various security measures in a bank management website.

- By enforcing strong password policies,
- encrypting passwords using AES-128, and
- adding additional authentication steps such as email-based one-time passwords (OTP) for modifying user details,

the website's security is significantly enhanced. These measures help prevent common security threats like

- brute force attacks,
- SQL injection,
- XSS, and
- unauthorized access, making the website more secure for users.

Modules

The bank management website is divided into several modules, each responsible for a specific functionality.

The modules include the user authentication module, which handles user login and registration, the password management module, which enforces strong password policies and encrypts passwords, the data modification module, which adds additional authentication measures for modifying user details,

and the security module, which protects the website from common security threats. Each module works together to ensure the security and integrity of the website's data and functionality.

Home page : It contains admin and user login link

Admin login page :While login we added SQL injection prevention technique. This page also responsible for sending OTP to user's email when user request to edit their data. Session hijacking is prevented.

User login page : While login we added sql injection prevention technique . This page also contains registration option if not registered. After registration password setting will be done during this we added strong password requirements . .Additional to this , we added authentication like permission request from admin. In this User login page we included the Cross Site Scripting prevention technique to prevent the attack which is affect the website. And also in user login system, we included the technique which encrypt passwords before storing them in the database to enhance security. This page also contains the prevention technique from the session hijacking attack is a type of attack where an attacker takes the user's active session to gain unauthorized access to a web Application.

Fractal:

Fractal Analytics is a pioneer in artificial intelligence, offering services across various sectors including consumer packaged goods, insurance, healthcare, retail, and technology ¹. Since its inception in 2000, Fractal has earned global recognition for its expertise in analytics and AI, helping Fortune 100 companies make informed decisions.

Key Facts About Fractal Analytics:

- *Global Presence:* Fractal has 15 global locations, including the United States, United Kingdom, Ukraine, and India ¹.
- *Leadership Recognition:* Featured as a leader on Forrester Wave: Customer Analytics Service Providers, 2017, and acknowledged as a “Hot Artificial Intelligence (AI)” company by Forbes ¹.
- *Awards and Recognition:* Distinctions as a “Cool Vendor” and a “Vendor to watch” by Gartner ¹.

Preparing for Your Interview:

To ace your interview, focus on:

- **Solidifying DSA Basics:** Ensure a thorough understanding of Data Structures and Algorithms (DSA) fundamentals ¹.
- **Effective Communication Skills:** Articulate your thoughts clearly and concisely ¹.
- **Comprehensive Resume Knowledge:** Be prepared to discuss every aspect of your resume in detail ¹.

Here are the values, mission, and vision of Fractal Analytics:

Values:

1. **Client Focus:** Deliver exceptional value to clients.
2. **Innovation:** Encourage experimentation and learning.
3. **Collaboration:** Foster open communication and teamwork.
4. **Integrity:** Act with transparency, ethics, and accountability.
5. **Respect:** Value diversity, inclusivity, and individuality.

Mission:

"Empower organizations to make better decisions through data-driven insights and AI-powered solutions."

Vision:

"To be the most respected analytics and AI company globally, driving transformative impact for clients."

Guiding Principles:

1. **Customer Centricity:** Prioritize client needs.
2. **Innovative Spirit:** Encourage experimentation.
3. **People First:** Foster growth, inclusivity, and well-being.
4. **Excellence:** Strive for quality, efficiency, and continuous improvement.

Leadership Philosophy:

1. **Empowerment:** Trust and empower teams.
2. **Transparency:** Foster open communication.
3. **Accountability:** Take ownership and responsibility.
4. **Continuous Learning:** Encourage growth and development.

Culture:

1. **Innovative:** Experiment, learn, and adapt.
2. **Collaborative:** Work together, share ideas.
3. **Diverse:** Embrace differences, promote inclusivity.
4. **Fun:** Enjoy the journey, celebrate successes.

Understanding these values, mission, and vision will help you:

1. Align with Fractal's culture and philosophy.
2. Showcase your fit during the interview process.
3. Make informed decisions about your potential role.

Feel confident and prepared to discuss Fractal's values and mission!

Here's an overview of the Imagineer role at Fractal Analytics:

Imagineer Role:

Imagineers at Fractal Analytics are innovative problem-solvers who combine design thinking, analytics, and technology expertise to drive business impact. They design and deliver cutting-edge solutions, leveraging AI, machine learning, and data science.

Key Responsibilities:

1. Design and develop innovative solutions using data science, AI, and machine learning.
2. Collaborate with cross-functional teams to identify business problems and develop solutions.
3. Develop prototypes and minimum viable products (MVPs) to test ideas.
4. Work closely with clients to understand their needs and deliver tailored solutions.
5. Stay up-to-date with industry trends and emerging technologies.

Skills Required:

1. Strong problem-solving skills and design thinking.
2. Proficiency in programming languages (Python, R, SQL).
3. Experience with data science tools (TensorFlow, PyTorch, scikit-learn).
4. Familiarity with cloud platforms (AWS, Azure, Google Cloud).
5. Excellent communication and storytelling skills.
6. Ability to work in agile environments.

Day-to-Day Activities:

1. Brainstorming sessions with teams to ideate solutions.
2. Developing and testing prototypes.
3. Collaborating with data engineers to integrate solutions.

4. Presenting solutions to clients and stakeholders.
5. Refining solutions based on feedback.